Year 12 Further Maths - Chi-squared tests

Similar questions - Goodness of fit

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A random sample of 60 observations is taken from a population which is thought to have the discrete uniform distribution on the integers $1, 2, \ldots, 12$. The observations are grouped as shown in the table.

Range	$1 \leqslant x \leqslant 4$	$5 \leqslant x \leqslant 8$	$9 \leqslant x \leqslant 12$
Observed frequency	18	y	$42 - y$

Find the range of integer values of y for which the data are not consistent with the distribution at the 5% significance level. **[9]**

2 In a standard model from genetic theory, the ratios of types a, b, c and d of a characteristic in a particular cross are predicted to be 9 : 3 : 3 : 1. Beth collects 200 specimens from such a cross and the numbers corresponding to each type of the characteristic are given in the table.

Type	a	b	c	d
Frequency	130	41	22	7

Beth tests, at the 1% significance level, whether the observed frequencies are consistent with the standard model.

(a) State appropriate hypotheses for the test. **[1]**

(b) Carry out the test. **[6]**

(c) State, with a reason, which one of the frequencies is least consistent with the standard model. **[1]**

(d) Suggest a different, improved model by changing exactly two of the ratio values. **[1]**

3 A teacher believes that when people are asked to choose an integer between 1 and 10, not all integers are equally likely to be chosen. She asks a random sample of 100 people to choose an integer between 1 and 10 inclusive. The observed frequencies O are given in the table, together with the values of $\dfrac{(O-E)^2}{E}$ where E is the corresponding expected frequency under the hypothesis that all integers are equally likely.

Integer	1	2	3	4	5	6	7	8	9	10
O	9	8	19	7	11	6	17	9	7	7
$\dfrac{(O-E)^2}{E}$	0.1	0.4	8.1	0.9	0.1	1.6		0.1	0.9	0.9

(a) Show that the missing value for the integer 7 is 4.9. **[2]**

(b) Show that there is evidence at the 5% significance level that the teacher's belief is correct. **[5]**

(c) The teacher suggests an alternative model: two of the integers each have probability p_1 of being chosen, and the other eight integers each have probability p_2. Suggest which two integers should have probability p_1 and a possible value of p_1. **[2]**

4 A machine tosses a coin 5 times. Pita believes that the outcomes of the tosses are not independent. She records the number of heads, X, in each of 80 sets of 5 tosses.

x	0	1	2	3	4	5	Total
Observed frequency	4	10	19	21	16	10	80
Expected frequency							
Contribution to χ^2	2.438	1.286	0.2012	0.8679			

(a) The sample mean and sample variance of X are 2.8125 and 1.852 respectively. Explain what these suggest about the validity of a binomial model $\mathrm{B}(5, p)$ for the data. **[3]**

Pita proposes a model in which the probabilities $\mathrm{P}(X = x)$, for $x = 0, 1, \ldots, 5$, are in the ratio 3 : 5 : 6 : 6 : 5 : 3. She carries out a χ^2 test to investigate whether this model is a good fit.

(b) Complete the table of expected frequencies and contributions. **[3]**

(c) Carry out the test using a 10% significance level. **[4]**

(d) Pita says that the results definitely show that her model is correct. Comment on this statement. **[2]**

Answers

Question 1

H_0: uniform distribution, so $E = 20$ in each group. $\chi^2 = \dfrac{(18-20)^2+(y-20)^2+(22-y)^2}{20}$. With $\nu = 2$ the 5% critical value is 5.991. Data inconsistent when $4+(y-20)^2+(22-y)^2 > 119.83$, i.e. $2y^2-84y+888 > 119.83$, i.e. $y^2-42y+384.09 > 0$. Roots $y = 21 \pm \sqrt{56.91} = 13.46,\ 28.54$. So the data are not consistent when $y \leqslant 13$ or $y \geqslant 29$ (consistent for $14 \leqslant y \leqslant 28$)

Question 2

(a) H_0: the frequencies fit the ratio 9 : 3 : 3 : 1; H_1: they do not

(b) Expected: 112.5, 37.5, 37.5, 12.5. Contributions: $\dfrac{17.5^2}{112.5} = 2.722$, $\dfrac{3.5^2}{37.5} = 0.327$, $\dfrac{15.5^2}{37.5} = 6.407$, $\dfrac{5.5^2}{12.5} = 2.42$. $\chi^2 = 11.88$. $\nu = 3$ and the 1% critical value is 11.345. $11.88 > 11.345$, so reject H_0: there is evidence at the 1% level that the data do not fit the standard model

(c) Type *c*: it has the largest contribution to χ^2 (6.407), the observed 22 being much lower than the expected 37.5

(d) For example 10 : 3 : 2 : 1 (changing the 9 and one 3): expected 125, 37.5, 25, 12.5 and $\chi^2 = 3.31$, which is not significant

Question 3

(a) $E = 100 \times \frac{1}{10} = 10$, so $\dfrac{(17-10)^2}{10} = 4.9$

(b) H_0: all integers are equally likely; H_1: not all equally likely. $\chi^2 = 0.1+0.4+8.1+0.9+0.1+1.6+4.9+0.1+0.9+0.9 = 18.0$. $\nu = 9$, 5% critical value = 16.919. $18.0 > 16.919$ so reject H_0: there is evidence at the 5% level that the teacher is correct

(c) Integers 3 and 7 (they have the largest observed frequencies). $p_1 \approx \dfrac{19+17}{200} = 0.18$ (and $p_2 \approx 0.08$)

Question 4

(a) For $B(5,p)$, $E(X) = 5p = 2.8125$ so $p = 0.5625$, and the variance would be $5p(1-p) = 1.23$. The sample variance 1.852 is much larger, suggesting the binomial model is not valid (the tosses are not independent)

(b) Expected $= 80 \times \dfrac{\text{ratio}}{28}$: 8.571, 14.286, 17.143, 17.143, 14.286, 8.571. Contributions: 2.438, 1.286, 0.201, 0.868, 0.206, 0.238

(c) H_0: ratio 3 : 5 : 6 : 6 : 5 : 3 is a good fit; H_1: not. $\chi^2 = 5.24$. $\nu = 6-1 = 5$; 10% critical value 9.236. $5.24 < 9.236$, so do not reject H_0: no evidence against the model

(d) Not rejecting H_0 does not prove the model is correct; the data are merely consistent with it (other models might fit equally well), and the binomial model has only been rejected in terms of its variance